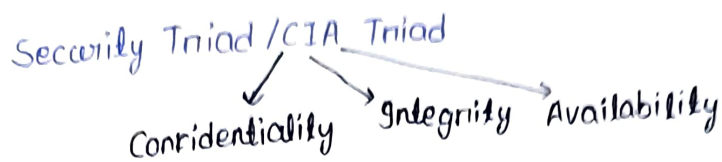


Section-1 Security Aims And Objectives

Chapter-1 Understanding Security Fundamentals



Computer security:-

- <1> The fundamentals of security are the foundation of protecting the user information.
- <2> The basic principle behind network security is the protection of the system and information from an unauthorized use.
- <3> Cryptography can reformat and transform user data to safeguard it. Computer security is a generic term related to tools required for protecting data and avoiding hackers.

Network Security:-

- <4> Network security are the features necessary for protecting data during their transmission.

Internet security are the features necessary to protect data during their transmission over a collection of interconnected network.

Security Attacks, Services and Mechanisms:-

- <5> Any action that compromises the security of information is a security attack. The mechanism that is designed to detect, prevent and recover from a security attack is the security mechanisms.

- <6> A service that enhances the security of data processing systems and information transfer is a security service. The common terms related to internet & network security are different security mechanisms such as

<A> Encipherment:- (Plain text $\rightarrow$ message)

This is the transformation of message or data with the help of mathematical algorithm and the most common techniques are called as cryptography and steganography.

 Digital Signature:- This is the method of electronic signing of data sender and electronic verification of the signature by the receiver.

<C> Data Integrity:- This refers to the method of ensuring the integrity or validity of the data by computing a numerical value associated with it.

<D> Authentication Exchange:- This refers to the exchange of information between two users providing their identity to each other.

<E> Routing Control and Access Control:- The section of secured route for data transfer is routing control and methods used to ensure that a user has the right to access the data on resource is access control.

<F> Principles of Security Services:-

The most popular principle for any security service is to follow the concept of CIA triad:-

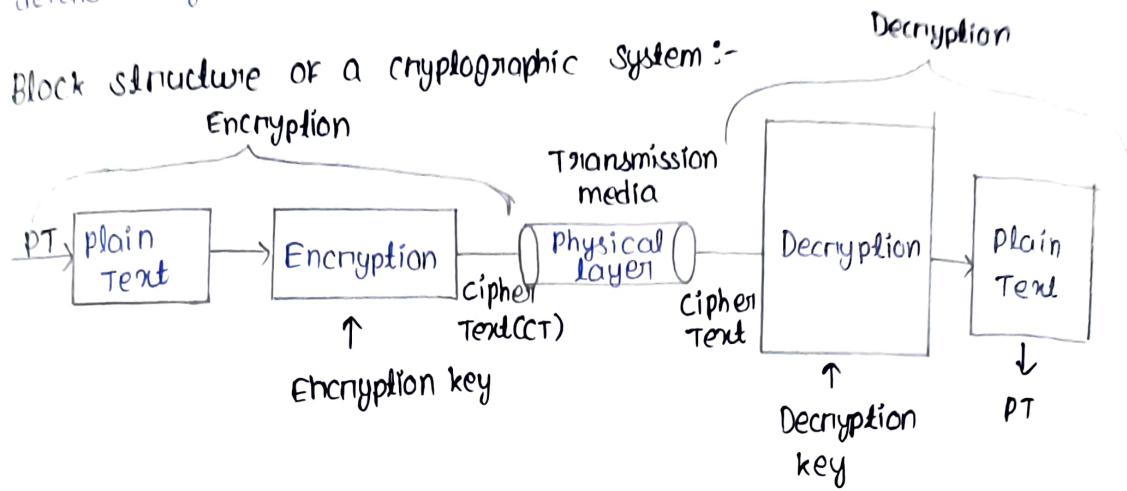
<a> Confidentiality:- It implies that the information is only accessible by the source or the receiver and thereby ensure that the users sensitive information remains shielded from an unauthorized access.

 Integrity:- It ensures that the users personal data should remain unaltered and trust worthy. Because data modification causes loss of integrity.

<C> Availability: This principle guarantees that the digital services are accessible whenever necessary without any problem.

All these ³ principles work in a harmony for creating a defensive system against security threats.

<8> Block structure of a cryptographic system:-



<9> There are 4 different categories of security control in a computer network.

<a> Technical:- It focuses to minimize the vulnerability in computer network by using data encryption and firewall concepts.

 Managerial:- It includes different planning and performance monitoring control for minimizing security risk in a network.

<c> Operational:- It includes different quality standards for enhancing productivity and optimising efficiency in a network.

<d> Physical:- It includes the range of techniques for preventing unauthorized access and examples are biometric lock and CCTV surveillance and tamper evident seal (digital watermarking).

<10> There exists different control types for ensuring the smooth operation of a network.

<a> Preventive Control:- It helps in minimizing potential threat to the system.

 Deterrent Control:- It helps in discouraging different attackers to engage in undesirable activity in system.

<C> Detective Control:- It helps in identifying & detecting security problems in a network.

<d> Connective Control:- It addresses different security problems.

<e> Compensating Control:- These are all alternative measures that helps in compromising with different technical issues.

<F> Directive Control:- This involves instructions and guidelines with the code or content to protect networking system.

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Chapter - 2

General Security Concepts:-

The fundamental security concepts in any networking system are the

- <1> CIA Triad (Confidentiality, Integrity & Availability)
- <2> AAA Concept (Authorization, Accountability & Authentication)

AAA Concept:-

AAA concept is the most important aspect of digital security.

Authentication:-

It is the process of identifying or verifying the user identity which may involve username, password or biometric information.

Authorization:-

It is the process of granting a user access to certain resources or permission to utilise network data or information.

Accountability:-

Accountability is the process of recording and monitoring a user activity on a system including time spent, resources used and actions taken.

This information can be used for real time monitoring of user action.

GAP Analysis:-

<1> It is the strategic process that evaluates the security practices in an organization by comparing it against the normal security standard and regulations.

<2> This assessment helps in identifying the difference or the gap between designed standards and acceptable standard.

<3> The major key points of gap analysis are:-

<a> Assessment:- To understand the organization's security policy and procedures.

- Benchmarking:- To compare existing security practices with respect to perfect security standard.
- <c> Identification:- To identify the label of security followed by a specific organization.
- <d> Prioritization:- To identify the security gaps based on priority, on potential impact.
- <e> Remediation Strategy:- The strategy outlines the steps required to improve the security practices in an organization.

What is the concept of Non-repudiation?

- <1> This concept prevents the denial of one's action ensuring proper accountability and reliability in electronic transaction and communication.
- <2> The key aspects of non-repudiation are digital signatures that confirms sender identity and integrity, audit trails that maintains a chronological record of user action, access control that helps in authentication of a user by granting permission to access information.
- <3> Non-repudiation helps in user-identification, authentication and authorization.

Zero trust:-

- <1> Zero trust in cyber security framework is based on the principle of never trust before verifying. It helps in preventing security breach by strictly utilising identity verification & continuous authentication.
- <2> It is based on the concept of data plane and security plane in a network architecture. Both planes are separated from each other with distinct responsibility.

Data plane:-

<1> It is responsible for actual movement and forwarding of data packets within a network, by focusing on executing tasks like routing, switching and data forwarding.

<2> Different entities in data plane initiate data communication while system represents a collective infrastructure for processing and traversing the data packet within the n/w.

<3> There are different types of zones in a system:-

<a> Implicit trust zone:-

These are the system areas where data communication happens without any verification.

 Internal Network zone:-

This is the local area network that can easily be controlled by Firewall system.

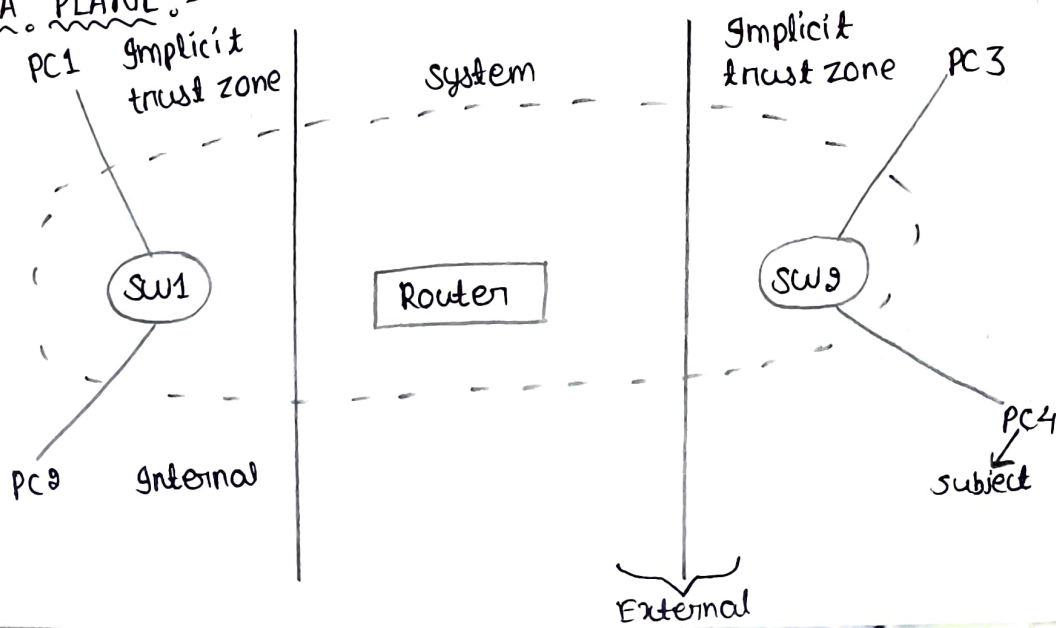
<c> DMZ (Demilitarized Zone):-

This area can't be trusted completely and works as a completely different secured network.

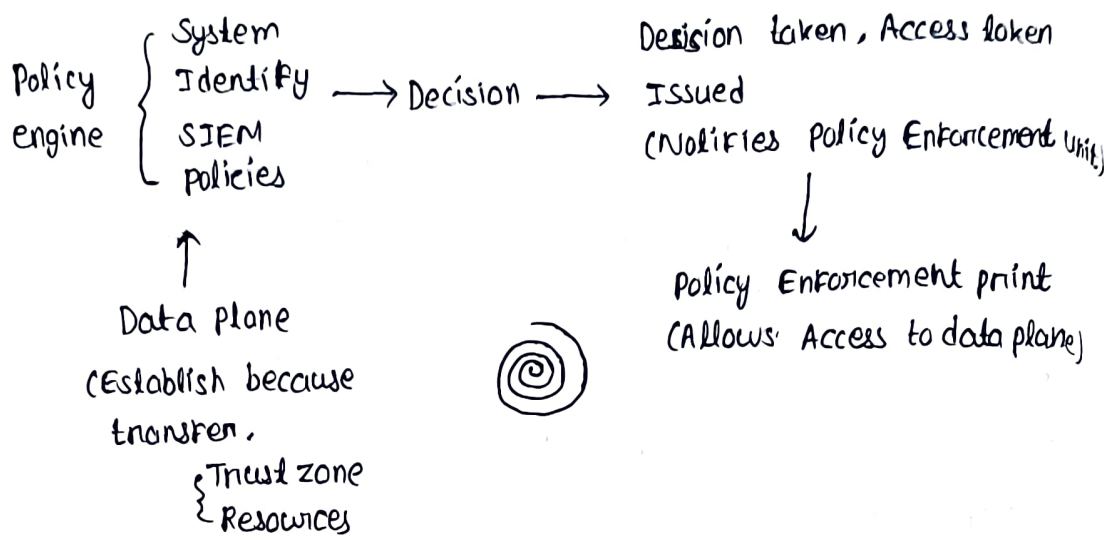
<d> External network zone:-

This area is completely untrusted requiring strong security criteria for authorization.

DATA PLANE:-



CONTROL PLANE:-



→ It manages the data acceptance, data routing, health of the network and device coordination.

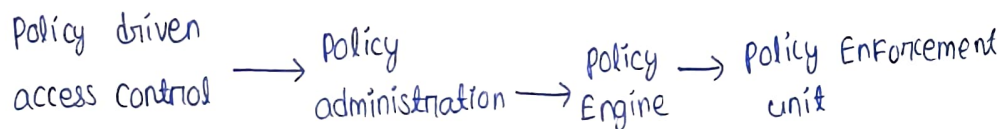
→ In a zero trust model it follows the principle of never trust and always verify validating the legitimacy of the user regardless of the location.

Working of Control Plane:-

→ This plane generates a centralised control for regulating access and activity monitoring.

→ The user is called as adaptive, identity which can change its behaviour, characteristics and location.

→ This plane also minimizes threat generated inside a networking by following a set of security policy and guideline step by step.



Physical security:-

→ It includes a range of measure required to prevent, detect & respond to potential risk in a computer network.

→ The different elements of physical security are:-

(1) Bollards, classic fencing, video surveillance, access badges, access control wristbands, sensor technologies that are based on infrared, microwave and ultrasonic rays.

Deception and Disruption Technology:-

This technology involves specific fake information for deceiving and disrupting potential threat in a network.

The common examples are:-

(a) Honeypot mechanism:- It is a cyber security tool that uses a reverse or decoy system for attracting cyber attacks to gain information about attackers and their techniques.

(b) Honeynet mechanism:- These are a group of honeypots in a network for safeguarding the network from malicious activities.

(c) Honeyfile mechanism:- It is a well designed original files with the password safe for attracting the attackers curiosity.

(d) Honey token mechanism:- It involves dummy data and dummy accounts to deceive the attacks.

(e) Fake information:- In this mechanism there exists a concept of DNS (sink hole) that is called as black hole of internet, where the DNS queries are deliberately redirected to different IP addresses.

Chapter-3 Importance of Change management and impact to Security

- <1> It is the process of making decisions about a network by using a combination of effective security policies and change monitoring.
- <2> The change management process is broken down into
- <a> Approval process
 - Ownership
 - <c> Invitation to stakeholders
- <3> Ownership refers to the person asking for inviting changes in a security network whereas stakeholder are groups or entities interested in the outcome of the change.
- <4> The key concept & process of change management are:-
impact analysis, test result, backout plan, maintenance window & standard operating procedures.
- <5> The technical implications of change management are the necessary operations that play an important role in maintaining digital harmony.

For example,

- Allow lists or white lists,
- Deny lists or black lists,
- Restricted activities &
- Documentation

- <6> Some factors impacting change management are minimization of downtime, minimization of time required for restarting or rebooting the users access to computing resources, applications and legacy application.

Chapter-5 Threat actors in a network

(1) Inside a network there exist different parameters, individual or group which are detrimental to the system, network and data. These are categorized as threat actors, hackers, cyber criminal.

(2) There exists 6 categories of threat actor which engage in different cyber criminal activities both benign and malicious.

→ The categories are:-

(i) Nation state threat actor

(ii) Advanced Persistent Threat (APT):- which is a sophisticated & focused cyber attack by well-funded organization.

(iii) Unskilled attacker:- Those who lack technical expertise, but can perform attack by purchasing tools from dark web.

(iv) Hacktivists:- They are individuals or groups that employ cyber tactics for promoting a cause, raise awareness or enact change.

(v) Inside the threats:- which originates from within an organization and are difficult to detect.

(vi) Organized crime:- where cyber criminals engage in activities like ransomware attacks, identity theft and credit card fraud.

(vii) Shadow IT:- It refers to the technology used within an organization without proper approval from the IT department. It creates vulnerabilities by exposing an organization to security risk.

Attributes of Threat Actors:-

→ The 3 critical attributes are:-

(i) Internal, external distinction, resource availability & level of sophistication. Internal threat originate from within an organization perform intentionally where as external threats are from outside the organization lacking direct knowledge about targets internal system.

The other attribute for performing the security attack is availability of proper resources and the last attribute is a highly sophisticated skilled attacker for performing attack.

→ The motivation that drives cyber attack is classified into following categories :-

<i> Financial gain

<ii> Data exfiltration that is data stealing

<iii> Espionage that implies gathering intelligence

<iv> Service disruption that destroys systems and services world wide

<v> Blackmailing or extortion using compromising information.

<vi> Political attacks initiated using digital activism.

<vii> Ethical attacks, disruption, revenge & cyber world.

Chapter-6Attack Surfaces

<1> The medium of attack introduced by the threat vector, is called attack surface. This includes unsecured network, both wired and wireless, vulnerable point like service port, phishing attack, impersonation, misinformation and social engineering.

<2> The most common message-based attack surfaces are

<a> Email :- EX :- Malicious attachments & phishing attack

 SMS :- It includes text-based scam, malicious links which are also called as phishing attack.

<c> Instant messaging (IM) :- which includes chat based phishing & malware distribution.

Image Based Attack :-

The malware information is hidden inside images and can lead to unauthorized ~~action~~ access, compromise of system and ransomware attack. The act of hiding information inside images is called as steganography.

File Based Attack :-

Files are like digital courier that can transform malicious information or Trojan viruses.

Voice call Based Attack :-

Attackers can manipulate voice call to deceive user into revealing personal information causing financial fraud or gaining unauthorized access. This type of attackers are called as vishing.

Removal device Attack:-

Devices like USB, an external harddisk can help in malware transmission.

Vulnerable software:-

Different software application with possible flaws can be exploited by malicious person for compromising data or disrupting system functioning. It is the task of anti virus system to perform vulnerable software scanning. Accordingly, 9 types of scanning are there.

<1> Client based Scanning
&

<9> Agentless Scanning

Unsupported Systems and Applications:-

Applications in unsupported system can use third-party software as prime target for introducing malicious information. An unsecured network is vulnerable to different security attacks by exploiting wifi & wired connection.

- <a> wireless connections can be hack by bluetooth or wifi.
- wired connection performs attacks on physically connected system.
- <c> Bluetooth can also exploit device connection.

Cyber Attack Through open service Port:-

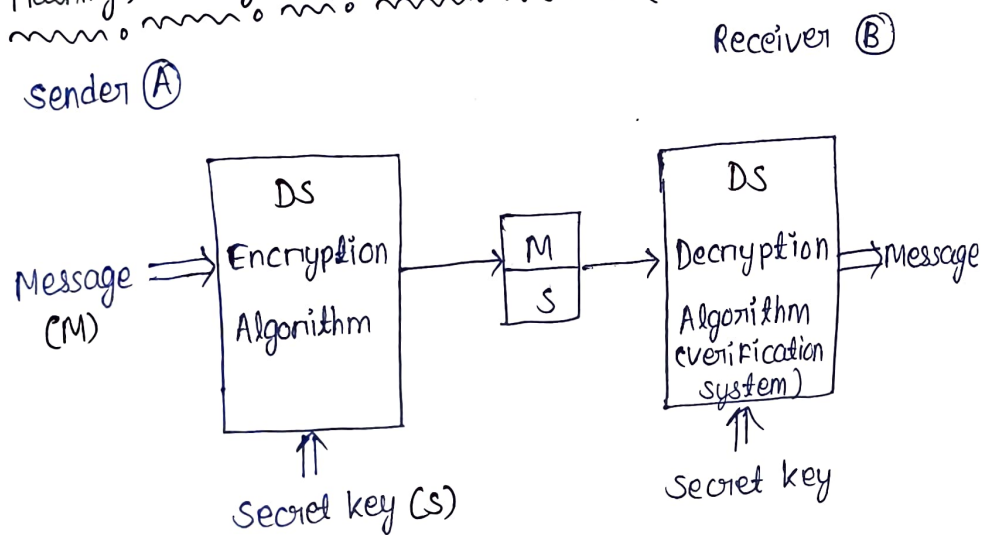
- <1> In this type of attack the hacker exploits open service port for unauthorized access and can sometimes generate cyber attack by utilising the default factory settings in a device.

⇒ Cyber attacks can also happen through third-party action provider or vendors, suppliers and MSP (Managed service providers).

(3) The human factors that are responsible for security breach inside a network and the study of this particular scenario is called as social engineering.

(4) The common factors are phishing, smishing, vishing, misinformation, impersonation, business email compromise, pretending which is fabricating scenarios to manipulate the user or the target, Typosquatting that is defined as exploitation of typo errors in domain name and watering hole concept that indicates the use of compromising websites for targeting attack.

Hashing, Salting and Digital Signature:-



DS → Digital Signature

(1) The diagram represents the working principle of digital Signature. DS is used for ensuring authenticity, integrity & non-repudiation principle in the document.

(2) The encryption happens in an asymmetric manner with private key & corresponding decryption or verification happens using public key.

<3> Hashing is a cryptographic process for maintaining data integrity & password security. Its purpose is to convert a data or password into a fixed size string by adding a salt value to & there after the hash value is computed from ^{it} which helps in authentication.

<4> Salting is a cryptographic technique that represent addition of a random string to a password before computing the hash value. This random string is called as salt.

Vulnerability in a security system is a type of fault or weakness inside the system, application or network that can be exploited by an outside agent for compromising the CIA concept. The different categories of vulnerability are :-

- <a> Application vulnerability :- It represents security fault in application due to
 - <i> Memory injection :- It is the authorized insertion of code into the system memory space.
 - <ii> Buffer overflow :- Here the data exceeds the allocated memory space.
 - <iii> Race condition :- It implies that conflicts will arise when multiple process will try to access the same shared resource.
 - <iv> Malicious updates :- The attackers will introduce harmful code into software during system update.
 - <v> SQL injection :- The attackers will manipulate the database ^{input} hacking the system.

 OS vulnerability :- It deals with different errors and faults happening at the operating system that includes

- <i> TOC (Time of Check) & TOU (Time of Use) :- These are exploited by the attacker where the program is unable to access resource as attacker have exploited the tiny gap between TOC & TOU.
- <ii> Cross site scripting :- It implies that the attacker introduces harmful code directly into the web page most commonly used by the user.

<C> Hardware vulnerability:- It implies the flaws in the physical component like CPU, Memory or peripherals that allow by passing the software security, leaking sensitive data or causing physical destruction. The examples are

i) Firmware:- Where the problems happen at low level system software that controls hardware.

ii) ^(old) Legacy:- These are older hardware with out dated security measures.

iii) End of life:- These are the security gaps introduced due to discontinued hardware support.

<d> Visualization vulnerability:- It involves graphical representation to identify and analyse security weaknesses in an IT environment. The examples are

i) VM escape:- It is a critical security breach where the user in a VM breaks through isolation barrier to interact with the host OS.

ii) VM sprawl:- It is uncontrolled proliferation of virtual machines in a virtualised environment causing significant wastage of resources.

iii) Resource Reuse:- This implies over use of shared resource causing system weakness.

<e> Cryptographic vulnerability:- These are the defects introduced in how the data is encrypted, stored or transmitted that can cause problems in confidentiality & integrity.

<f> Misconfiguration vulnerability:- These are the system errors introduced during system setup causing security loopholes.

(g) Zero day vulnerability:- These are the software or hardware security flaws unknown to the vendor exposing the system to security attacks.

(h) Supply chain vulnerability:- These are the weaknesses from the supplier side that exposes the system or organization to data disruption and financial loss. It includes service provider where the problems are in the third party supply chain, hardware & software provider where the problems are in the hardware & software sites.

(i) Mobile device vulnerability:- These include the defects in android and ios devices causing data leakage & critical device insecurities. The examples are side loading that is installing applications from unknown resources and Jail breaking that implies compromising device security by bypassing system ~~and~~ standards.

Indicators of Malicious

Chapter - 8

Attacks / Activity

Dt: 23/08/26

<1> Malware:-

It is a malicious software or code that is specifically designed to disrupt or damage system by gaining unauthorized access into it. It is categorized as:-

<a> PUPs (Potentially Unwanted Program):- These are

downloaded ^{along} with other programs consuming system resources & slowing down the system.

 Ransomware:- It encrypts private files demanding payment for release of it.

<c> Trojan:- These are PE (Portable executable) files containing executable codes hampering the system.

<d> Worms:- These malware can self propagate and stay within computer memory with an ability to replicate.

<e> RATs:- It is remote access trojan that are hidden invaders which can compromise system safety by running different executable files.

<f> Spyware:- which can slow down system activity by hacking into user credentials.

<g> Bloatware:- These malware can slowdown the system performance by excessive resource consumption.

<h> Virus:- These are impacted file or software that can replicate within system memory & a category of virus that can modify system SW called polymorphic virus.

<i> Key loggers:- These are silent digital software observers that can record the strokes on the keyboard capturing credentials like password or credit card

details.

<3> Logic Bombs and Rootkits:- These are the kind of software that can trigger unnecessary events like system crash. Rootkits are software that can hide their presence deep within the OS.

***<9> Network Attacks:-

The network attack is an unauthorized attempt to compromise or gain access into a complete network. It is categorized as:

- <a> Pivoting
- DDOS (Distributed denial of service)
- <c> DNS attack

<a> Pivoting:- This attack is implemented by targeting critical servers in a network like database servers or domain controllers.

*** DDOS:- This is a type of attack where the host prevents the network services from operating. This is activated from multiple hacker websites and they can control the complete systems in a network. The hacker systems are called as Botnets, and they can generate attack divided into 2 categories.

<1> Network amplified attack where the complete network communication is destroyed.

<2> Network reflected attack where the attacker can extract the victim ip addresses through the third party server destroying the system.

<c> DNS attack:- Domain Name System attacks are cyber attacks that can disrupt system functionality leading the system or redirecting a ^{to} different hacker websites causing data theft.

and other harmful consequences.
→ While a user specific URL, it uses DNS name resolution find the corresponding IP. In this case the attacker can attach different IP addresses through websites causing DNS attack.

→ The key concept of DNS attack are:-

<i>DNS Name Resolution

DNS cache



host files



*** Root hints

The most common websites that are regularly browsed are keeping DNS cache memory as common URLs. If the URL is not available in the cache memory than the system look for it in the host file (Uniform Resource Locator). In the absence of URL located on host file the DNS server will forward the request to other near by DNS servers using

Root hints.

<ii> DNS Sinkhole:- The task of sinkhole is to identify the fake domain names preventing potential hackers on attackers from entering into system.

<iii> DNS cache poisoning:- It occurs when an external attacker manipulates the DNS records redirecting user with fake information.

<3> Wireless Attacks:-

These attacks are generated usually at wireless access point using 9 methods.

<A> Rogue Access Points (RAP):-

RAP pretends to be a legitimate wireless access point that helps in tricking users into connecting and sharing sensitive information.

 Evil Twins:-

It takes the concept of RAP to another level and intercepts communication between users and networks by creating duplicate network with SSID similar to original network.

<4> On Path Attacks:-

These are called as interception attack or man in the middle attack that intercepts communication between two parties by acting as silent cyber criminal exploiting sensitive information and modifying it.

It is of 2 categories:-

<a> Session Replay Attacks:- The attack is generated by hijacking a specific session exploiting information.

 Replay Attacks:- This is an unpath attack that resends / replace ^{the} data continuously at later times.

<a> Substitution technique :- (Replacement)

This techniques are where the characters are replaced by other symbols or characters using a specific mathematical model & a secret key.

Ex:- ABC ABC
 ↓
 (EFG) easily encrypted (% - +) → not encrypted
 → Secret key → 5

 Transposition technique :- (Rearrangement)

In transposition technique, we apply a type of permutation to generate a scrambled plain text called as cipher.

Ex:- ABC
 → CBA
 → BAC
 → ACB

Substitution technique :-

Depending upon the type of mathematical concept the techniques are categorized as

<a> Caesar Cipher :-

It is a simple substitution cipher where every letter in the plain text are shifted by Fixed number of characters down the series.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

$$CT = E(PT, k) = (PT + k) \bmod 26$$

$$PT = D(CT, k) = (CT - k) \bmod 26$$

{ key = 3 }

Q PT = ZEBRA
 ↓ ↓ ↓ ↓ ↓
 25 4 1 17 0

+3
 28 7 4 20 3
 mod 26 9 7 4 20 3
 ↓ ↓ ↓ ↓ ↓

ZEBRA ⇒ CHEUD

- (i) Encrypt APPLE using Caesar cipher.
 (ii) Decrypt FRPSXWHU QZ using Caesar cipher.

(i) APPLE
 ↓ ↓ ↓ ↓ ↓
 0 15 15 11 4

+3
 3 18 18 14 7
 mod 26 3 18 18 14 7
 D S S O H

APPLE $\Rightarrow$ DSSOH

(ii) FRPSXWHU QZ
 ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
 5 17 15 18 23 22 7 20 16 25

-3
 2 14 12 15 20 19 4 17 13 22
 mod 26 2 14 12 15 20 19 4 17 13 22
 C O M P U T E R N W

FRPSXWHU QZ $\rightarrow$ COMPUTER NW

$\rightarrow$ Although Caesar cipher is easy to implement & it is the simplest technique, but the disadvantage is that its encryption and decryption algorithms are similar for every value of secret key and we can have only 25 different keys available. Therefore, any cipher text (CT) can be broken down in 25 attempts at least. So, this is not a secure cipher.

(b) Monalphabetic Cipher:-

$\rightarrow$ It is a single cipher substitution technique where every plain text character maps to a fixed unique letter or symbol throughout the entire message.
 $\rightarrow$ Depending on the provided PT, CT mapping, we have to generate an alphabetic series & use it for calculating the cipher text.

Encryption

$\rightarrow$ For example, APPLE $\rightarrow$ 10094

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	0	9	E	4	%	-	+	X	F	x	q	y	z	Q	o

$\rightarrow$ It's advantage is that it is simple to implement & understand.
 But its disadvantage is that it is highly vulnerable to Frequency analysis and the cipher can be broken down with the

help of ~~26 factorial~~ $26! \approx 4 \times 10^{26}$ different keys provided

every character is mapped to another character in the alphabetic series. It does not work for ^{common patterns containing} double letters (Disadvantage) as key.

Difference between Caesar cipher & Monoalphabetic cipher:

→ Monoalphabetic is a substitution cipher where every letter is repeated by another but the key is not defined.

The encryption & decryption is performed by generating a map between PT & CT

Q1 Find out the corresponding monoalphabetic cipher if the key is Monarchy & the PT is secret.

Ans:- Extended key = MONARCHY BDEFGHIJKLPQRSTUVWXYZ
ABCDEF GHIJKLMNOPQRSTUVWXYZ

secret $\Rightarrow$ QRNPRS

Q2 Find out the monoalphabetic cipher corresponding to secret key Network if plain text is cyberwar.

Ans:- Extended key = NETWORK ABCDEFGHIJLMNOPQRSTUVWXYZ
ABCDEFGHIJKLMNOPQRSTUVWXYZ

~~secret~~ Cyberwar $\Rightarrow$ TYEOMVNM

Q3 Find out the monoalphabetic cipher or key mapping is:-

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Q W E R T Y U I O P A S D F G H J K L Z X C V B N M

Plain Text :- HELLO

→ ITSSG (Ans)

<c> Vignere cipher :- (Polyalphabetic cipher)

- It is a poly alphabetic cipher that uses multiple Caesar cipher based on letters of a keyword.
- A keyword controls the encryption and every letter of the key indicates how many positions should the plain text letter should be shifted.

Q Find out the Vignere cipher for PT = Attackdown and

key = Lemon.

Ans:-
PT = A t t a c k d a w n
CT = L E M O N L E M O N
① 19 19 0 2 10 ③ 0 22 13
① 4 19 14 13 11 ④ 19 14 13
L X F O P H M K A
11 23 5 14 15 21 7 12 10 0

Advantages :-

- It is more secure than Caesar cipher. Because it uses multiple alphabets for shifting the PT.

Disadvantages :-

- If the length of the secret key is small then it can be easily decrypted.

$$CT = (PT + k) \bmod 26 \text{ (Encryption)}$$

$$PT = (CT - k) \bmod 26 \text{ (Decryption)}$$

<d> Vernam cipher :- (PC)

- It is a symmetric ciphering in technique where the encryption is performed by XOR operation between plain text & secret key.

$$CT = (PT \oplus k) \bmod 26$$

$$PT = (CT \oplus k) \bmod 26$$

- This cipher is also called as one-time pad technique whenever the length of the key is same as the PT and the key is completely random.

→ It's advantage is that due to XOR operation, it is simple to implement with faster encryption but the disadvantage is that the same operation is used for both encryption & decryption and whenever the key length is small this cipher is breakable.

Q Encrypt PT = Hello
key = key

A²

H	E	L	L	O
7	4	11	11	14
⊕ K	⊕ E	⊕ Y	⊕ K	⊕ E
10	4	24	10	4

HELLO $\Rightarrow$ NATBK

	16	8	4	2	1
	0	0	1	1	1 (H)
⊕	0	1	0	1	0 (K)
	0	1	1	0	1 = 13 (N)
	E	⊕	E		= 0 (A)
	16	8	4	2	1
	0	1	0	1	1 (11) / (L)
⊕	1	1	0	0	0 (24) / (Y)
	1	0	0	1	1 → 19 (T)
	0	1	0	1	1 (11 → L)
⊕	0	1	0	1	0 (10 → K)
	0	0	0	0	1 = 1 (B)
	0	1	1	1	0 (0)
⊕	0	0	1	0	0 (E)
	0	1	0	1	0 → 10 (K)

<2> PT = Dog
key = cat

D	O	G
3	14	6
C	A	T
2	0	19

Dog $\Rightarrow$ BOV

	0	0	0	1	1 (D)
⊕	0	0	0	1	0 (C)
	0	0	0	0	1 → 1 (B)
	0	1	1	1	0 (0)
⊕	0	0	0	0	0 (A)
	0	1	1	1	0 → 14 (O)
	0	0	1	1	0 (G)
⊕	1	0	0	1	1 (T)
	1	0	1	0	1 → 21 (V)

<e> Playfair Cipher :- (cc)

→ It is a bi-graph substitution cipher which is much more secure compare to previous cipher in techniques.

→ It encrypts a pair of letters at once instead of single letter by using a (5×5) grid matrix constructed from secret key. The grid excludes the letter J that is also sometimes combine with letter I.

→ Every pair of PT letter is encrypted based on there positions in the ~~there~~ grid.

<a> If the pair of letter is present in the same row then for encryption we have to take the letter to the immediate right and for decryption we have to take the letters to the immediate left in a circular manner.

 If the pair of letters are in the same column then for encryption we have to take the letters in a downward manner & for decryption in an upward manner circularly.

<c> If the pair of letters are not in the same row & column then for a rectangle by taking the letters and consider the letters on the same row but on opposite corner of the rectangle for both encryption and decryption.

Q Encrypt BALONX using key = Monarchy.

PT = BA LO NX

Ba lo nx

IB PM AW

M	O	n	a	r
c	h	y	b	d
e	f	g	i/j	k
l	p	q	s	t
u	v	w	x	z

BALONX $\Rightarrow$ IBPMAW

Q PT = Balloon

key = Monarchy

PT = Ba l l o o n
Ba l x lo o n

CT = IBSUPMNA

M	O	N	A	1
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

<P> Hill Cipher:- (PC)

→ It is a polyalphabetic substitution cipher based on the concepts of linear algebra.

→ Every algebraic computation is represented as modulo 26 operation.

→ The PT is divided into characters based on the size of the secret key and the secret key is mandatory to have an ~~inv~~ invertible matrix format.

→ Mathematically encryption is represented as

$$C = KP \pmod{26}$$

and decryption is represented as

$$P = K^{-1}C \pmod{26}$$

$$A^{-1} = \frac{1}{|A|}(\text{adj}A)$$

$$= (K^{-1} \pmod{26}) (C \pmod{26})$$

$$= \left(\frac{1}{|K|} \pmod{26} \right) ([\text{adj}.K] \pmod{26}) (C \pmod{26})$$

$$= (K \text{ inv} \pmod{26}) ([\text{adj}.K] \pmod{26}) (C \pmod{26})$$

Encryption:-

$$K = \begin{bmatrix} 2 & 3 \\ 9 & 3 \end{bmatrix}$$

It is not invertible.

not possible.

1. Encrypt "Attack" using $k = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$

Ans: $|k| = (2 \times 6 - 3 \times 3) = 3$ (Hill encryption possible)

$$C = kP \text{ mod } 26$$

$\swarrow \quad \searrow$
 $2 \times 2 \quad 2 \times ?$

$$P = \text{Attack} \quad \begin{matrix} P_1 & P_2 & P_3 \\ \begin{pmatrix} A \\ T \end{pmatrix} & \begin{pmatrix} T \\ A \end{pmatrix} & \begin{pmatrix} C \\ K \end{pmatrix} \end{matrix}$$
$$= \begin{pmatrix} 0 \\ 19 \end{pmatrix} \begin{pmatrix} 19 \\ 0 \end{pmatrix} \begin{pmatrix} 2 \\ 10 \end{pmatrix}$$

$$C_1 = kP_1 \text{ mod } 26$$

$$= \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 19 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 2 \times 0 + 3 \times 19 \\ 3 \times 0 + 6 \times 19 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 57 \text{ mod } 26 \\ 114 \text{ mod } 26 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 10 \end{bmatrix} = \begin{bmatrix} F \\ K \end{bmatrix}$$

$$C_2 = kP_2 \text{ mod } 26$$

$$= \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 19 \\ 0 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 2 \times 19 + 3 \times 0 \\ 3 \times 19 + 6 \times 0 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 38 \text{ mod } 26 \\ 57 \text{ mod } 26 \end{bmatrix}$$

$$= \begin{bmatrix} 12 \\ 5 \end{bmatrix} = \begin{bmatrix} M \\ F \end{bmatrix}$$

$$C_3 = kP_3 \text{ mod } 26$$

$$= \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 2 \times 2 + 3 \times 10 \\ 3 \times 2 + 6 \times 10 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 34 \text{ mod } 26 \\ 66 \text{ mod } 26 \end{bmatrix}$$

$$= \begin{bmatrix} 8 \\ 14 \end{bmatrix} = \begin{bmatrix} I \\ O \end{bmatrix}$$

Attack $\xrightleftharpoons[\text{Decrypt}]{\text{Encrypt}}$ FK MF IO

2. Encrypt using hill cipher in PT = HILL and $k = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

Ans: $|k| = (3 \times 5 - 3 \times 2) = 9$ (it is possible).

$$C = kP \pmod{26}$$

$$\downarrow \quad \downarrow$$
$$2 \times 2 \quad 2 \times 2$$

$$PT = \underline{HILL} \quad \begin{matrix} P_1 & P_2 \\ \begin{pmatrix} H \\ I \end{pmatrix} & \begin{pmatrix} L \\ L \end{pmatrix} \end{matrix}$$
$$= \begin{pmatrix} 7 \\ 8 \end{pmatrix} \begin{pmatrix} 11 \\ 11 \end{pmatrix}$$

$$C_1 = kP_1 \pmod{26}$$

$$= \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 3 \times 7 + 3 \times 8 \\ 2 \times 7 + 5 \times 8 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 45 \pmod{26} \\ 54 \pmod{26} \end{bmatrix}$$

$$= \begin{bmatrix} 19 \\ 9 \end{bmatrix} = \begin{bmatrix} T \\ C \end{bmatrix}$$

$$Hill \xrightleftharpoons[D]{E} TCOZ$$

$$C_2 = kP_2 \pmod{26}$$

$$= \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 11 \\ 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 3 \times 11 + 3 \times 11 \\ 2 \times 11 + 5 \times 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 66 \pmod{26} \\ 77 \pmod{26} \end{bmatrix}$$

$$= \begin{bmatrix} 14 \\ 25 \end{bmatrix} = \begin{bmatrix} O \\ Z \end{bmatrix}$$

Decryption:

→ For decrypting we require information about inverse modulo operator and negative value operator.

$$a \pmod{b} = \text{rem}(a/b)$$

$$-a \pmod{b} = (b - a \pmod{b})$$

Ex:

$$9 \pmod{5} = 4$$

$$-9 \pmod{5} = (5 - 9 \pmod{5})$$
$$= (5 - 4) = 1$$

$$-17 \pmod{26} = (26 - 17 \pmod{26})$$
$$= (26 - 17) = 9$$

$$-28 \pmod{26} = (26 - 28 \pmod{26})$$
$$= (26 - 2) = 24$$

Inverse modulo operation:

1) (a) 2 inv mod 9 = 5
 $2 \times 1 = 2 \mod 9 = 2 \neq 1$
 $2 \times 2 = 4 \mod 9 = 4 \neq 1$
 $2 \times 3 = 6 \mod 9 = 6 \neq 1$
 $2 \times 4 = 8 \mod 9 = 8 \neq 1$
 $2 \times 5 = 10 \mod 9 = 1$

(c) 3 inv mod 44 = 15
 $3 \times 14 = 42 \mod 44 = 42 \neq 1$
 $3 \times 15 = 45 \mod 44 = 1$

(b) 11 inv mod 21 = 2
 $11 \times 11 = 121 \mod 21 = 121 \neq 1$
 $11 \times 2 = 22 \mod 21 = 1$

(d) 9 inv mod 26 = 3
 $9 \times 3 = 27 \mod 26 = 1$

(e) 17 inv mod 26 = 23
 $17 \times 23 = 391 \mod 26 = 1$

2) Compute the inv mod operation for the following matrices.

(a) $\begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$ inv mod 11

Det = $(12 - 9) = 3$
 $6 \text{ inv mod } 11 = 2$
 $3 \times 2 = 6 \mod 11 = 6 \neq 1$

Det = 3

$A^{-1} = \left(\frac{1}{|A|} \mod 11 \right) (\text{adj. } A \mod 11)$
 $= (3 \text{ inv mod } 11) \left(\begin{bmatrix} 6 & -3 \\ -3 & 2 \end{bmatrix} \mod 11 \right)$

$= 4 \begin{pmatrix} 6 & -3 \\ -3 & 2 \end{pmatrix} \mod 11$

$= \begin{pmatrix} 24 & -12 \\ -12 & 8 \end{pmatrix} \mod 11$

$= \begin{pmatrix} 2 & 10 \\ 10 & 8 \end{pmatrix}$

(b) $\begin{bmatrix} 2 & 8 \\ 3 & 4 \end{bmatrix}$ inv mod 21

Det = $(8 - 6) = 2$

$A^{-1} = \left(\frac{1}{|A|} \mod 21 \right) (\text{adj. } A \mod 21)$
 $= (2 \text{ inv mod } 21) \left(\begin{bmatrix} 4 & -3 \\ -2 & 2 \end{bmatrix} \mod 21 \right)$

$= 11 \begin{pmatrix} 4 & -3 \\ -2 & 2 \end{pmatrix} \mod 21$

$= \begin{pmatrix} 44 & -33 \\ -22 & 22 \end{pmatrix} \mod 21 = \begin{pmatrix} 2 & 9 \\ 9 & 1 \end{pmatrix}$

Q Decrypt $C = HIAT$, if $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$.

Ans: $C_1 = \begin{bmatrix} H \\ I \end{bmatrix}$ $C_2 = \begin{bmatrix} A \\ T \end{bmatrix}$ $|K| = (15-6) = 9$

$$P = K^{-1}C \pmod{26}$$

$$P_1 = \left(\frac{1}{|K|} \pmod{26} \right) (\text{adj. } K \pmod{26}) (C_1 \pmod{26})$$

$$= (9 \text{ inv mod } 26) \left(\begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} \pmod{26} \right) \left(\begin{pmatrix} 7 \\ 8 \end{pmatrix} \pmod{26} \right)$$

$$= \left(3 \begin{pmatrix} 5 & -3 \\ -2 & 3 \end{pmatrix} \pmod{26} \right) \left(\begin{pmatrix} 7 \\ 8 \end{pmatrix} \pmod{26} \right)$$

$$= \left(\begin{bmatrix} 15 & -9 \\ -6 & 9 \end{bmatrix} \pmod{26} \right) \left(\begin{pmatrix} 7 \\ 8 \end{pmatrix} \pmod{26} \right)$$

$$= \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 15 \times 7 + 17 \times 8 \\ 20 \times 7 + 9 \times 8 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 105 + 136 \\ 140 + 72 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 241 \pmod{26} \\ 212 \pmod{26} \end{bmatrix}$$

$$= \begin{bmatrix} 7 \\ 4 \end{bmatrix} = \begin{bmatrix} H \\ E \end{bmatrix}$$

$$P_2 = \left(\frac{1}{|K|} \pmod{26} \right) (\text{adj. } K \pmod{26}) (C_2 \pmod{26})$$

$$= (9 \text{ inv mod } 26) \left(\begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} \pmod{26} \right) \left(\begin{pmatrix} 0 \\ 19 \end{pmatrix} \pmod{26} \right)$$

$$= \left(\begin{bmatrix} 15 & -9 \\ -6 & 9 \end{bmatrix} \pmod{26} \right) \left(\begin{pmatrix} 0 \\ 19 \end{pmatrix} \pmod{26} \right)$$

$$= \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 19 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 323 \text{ mod } 26 \\ 171 \text{ mod } 26 \end{bmatrix}$$

$$= \begin{bmatrix} 11 \\ 15 \end{bmatrix} = \begin{bmatrix} L \\ P \end{bmatrix}$$

Q. $C = AVCA TY$

$$K = \begin{bmatrix} 3 & 3 \\ 9 & 5 \end{bmatrix}$$

calculate PT.

$$|K| = (15 - 6) = 9$$

Ans.

$$C_1 = \begin{bmatrix} A \\ V \end{bmatrix} \quad C_2 = \begin{bmatrix} C \\ A \end{bmatrix} \quad C_3 = \begin{bmatrix} T \\ Y \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 21 \end{bmatrix} \quad = \begin{bmatrix} 9 \\ 0 \end{bmatrix} \quad = \begin{bmatrix} 19 \\ 24 \end{bmatrix}$$

$$P_1 = (9 \text{ inv mod } 26) \left(\begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} \text{ mod } 26 \right) \left(\begin{bmatrix} 0 \\ 21 \end{bmatrix} \text{ mod } 26 \right)$$

$$= \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 21 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 357 \text{ mod } 26 \\ 189 \text{ mod } 26 \end{bmatrix}$$

$$= \begin{bmatrix} 19 \\ 7 \end{bmatrix} = \begin{bmatrix} T \\ H \end{bmatrix}$$

$$P_2 = (9 \text{ inv mod } 26) \left(\begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} \text{ mod } 26 \right) \left(\begin{bmatrix} 9 \\ 0 \end{bmatrix} \text{ mod } 26 \right)$$

$$= \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 9 \\ 0 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 30 \text{ mod } 26 \\ 40 \text{ mod } 26 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 14 \end{bmatrix} = \begin{bmatrix} E \\ O \end{bmatrix}$$

$$P_3 = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 19 \\ 24 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 285 + 408 \\ 380 + 216 \end{bmatrix} \text{ mod } 26 = \begin{bmatrix} 693 \text{ mod } 26 \\ 596 \text{ mod } 26 \end{bmatrix} = \begin{bmatrix} 17 \\ 24 \end{bmatrix} = \begin{bmatrix} R \\ Y \end{bmatrix}$$

$\Rightarrow$ PT = THEORY

Advantages and Disadvantages of Hill cipher :-

- <1> Unlike monoalphabetic cipher, hill cipher encrypts multiple letters at once. This makes the frequency analysis much harder. This encryption of multiple letters at once is called as a block cipher and its opposite is stream cipher.
- <2> In hill cipher repeating letters of plain text gets mapped to different cipher text (CT) letters improving security.
- <3> Due to matrix multiplication its computation becomes difficult for larger matrix.
- <4> The size of the secret key is flexible and larger matrices implies stronger encryption.
- <5> The disadvantage is that the secret key must be a square matrix and should be invertible in nature otherwise decryption is impossible.
- <6> It is a classical cipher, but can easily be broken with modern computing.
- <7> The size of the PT has to match the size of the secret key in every ~~code~~ cipher.

Transposition ciphers :-

→ The ciphertext is generated by applying permutation and combination over the PT.

→ It is classified into 2 types :-

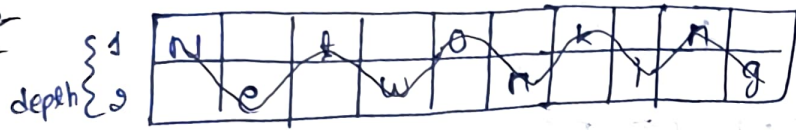
- <a> Rail Fence Technique
- Row column Transposition Technique

(a) Rail Fence Technique :-

Here the PT is ~~read~~ rewritten as a sequence of diagonal element which is there after read and sequence of rows. The diagonal formation depends on the provided secret key which is the depth.

Q Encrypt "Networking" using rail fence if the key = 2.

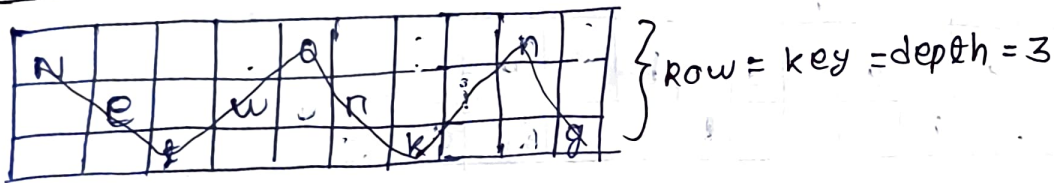
Ans:-



key = 2
Row = 2

CT = NTOKNEWRIg

For key = 3



CT = NONEWRITKG

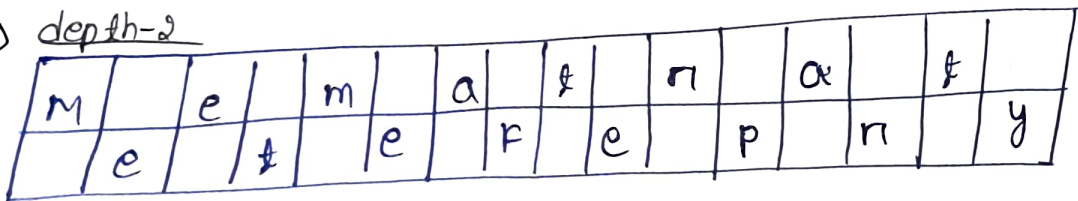
* Does not consider space. Ex:- COMPUTER NETWORK.

Q Perform Rail Fence Encryption.

<i> PT = Meet me after party. (depth = 2 & 3)

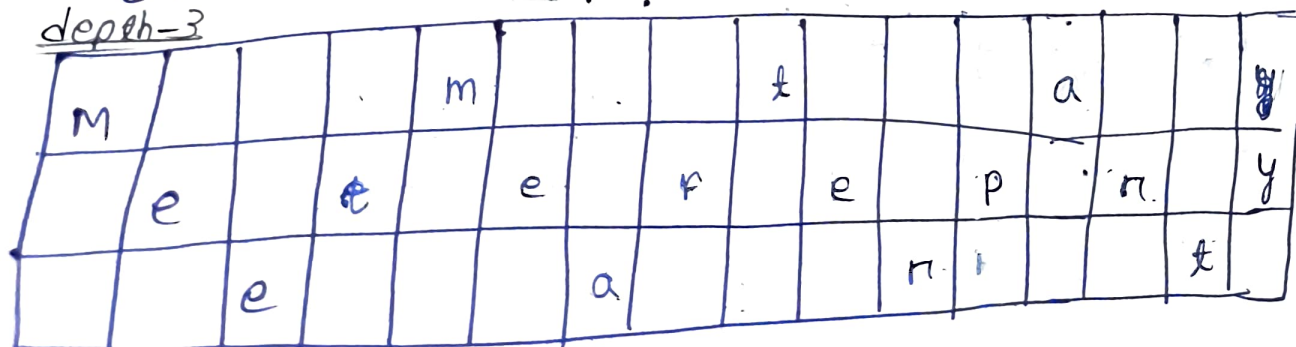
<ii> PT = Defend the East. (depth = 3 & 2)

Ans:- <i> depth-2



CT = MEMATRA TETEFEPY

depth-3



CT = MMTAETEFEPRYEART

(ii)

depth = 2

D	f	n	t	e	a	t
e	e	d	h	e	s	

CT = DDFNTEATEEDHES

depth = 3

D			n			e				t
	e		e		d	h		e	s	
		r			t			a		

CT = DNETEEDHESFTA

Q Decrypt "Defend the East", depth = 3.

D			n			e				t
	e		e		d	h		e	s	
		r			t			a		

CT = DNETEEDHESFTA

Q Decrypt using rail fence for depth of 2

CT = G M O E A E V R

Ans

G		M		O		E
	A		E		V	
						R

GAMEOVER

depth = 3

CT = DNETEEDHESFTA

D			n			e				t
	e		e		d	h		e	s	
		r			t			a		

PT = DEFENDTHEEAST

→ Rail Fence technique is easy to implement but the key is known it can easily be decrypted and the no. of letters are also similar in both PT & the CT.

→ A better technique is the row column transposition.

(b) Row Column Transposition Technique:-

→ It is a transposition cipher where the PT are rearrange based on grid for using a key word.

→ The simple rearrangement of both PT & CT is depending on the alphabetical placement of the secret key.

→ The size of the grid / matrix is depending on the secret key where the no. of columns is equivalent to the length of the key.

Q1 Encrypt using "Row column Technique" where key = Zebra and PT = "We are discovered run".

Ans:- key = zebra (length = 5)

P.T = "We are discovered run"

Z	E	B	R	A
5	3	2	4	1
w	e	a	r	e
d	i	s	c	o
v	e	n	e	d
r	u	n	x	x

← x5

CT = EODXASRNEIEURCEXWDVR

Q.2 Encrypt PT = Attack at dawn, key = Hack

Ans:-

H	A	C	K
3	1	9	4
A	t	t	a
c	k	a	t
d	a	w	n

3x4

CT = TKATAWACDATN

 PT = Hello world, key = key.

Ans:-

K	E	Y
2	1	3
H	e	l
l	o	w
o	n	l
d	X	X

4x3

CT = EORXHL0DLWLX

Q.3 Decrypt the cipher text = HTHES THHR NDAE IAE0ETT,

key = Earth.

Ans:-

E	a	r	t	h
9	1	4	5	3
S	H	E	O	R
T	T	I	E	N
H	H	A	T	D
H	E	E	T	A

PT = SHEORTTJENHHATDHEETA

Q.4

Block Cipher vs Stream Cipher

PT = Computer Network

PT = COMPUTER NETWORK

CT = COM NET

CT = DPNQVUESOFUXPSL

→ In stream cipher the representation is in sequential format where every bit is responsible for encryption & decryption.

Ex:- All substitution & transposition cipher.

→ In block cipher the encryption & decryption happens for individual block and the resultant cipher text length may not be similar to that of PT. The encryption is block cipher is based on the concept of confusion & diffusion of cryptography.

Ex:- DES & AES

DES:-

(Data Encryption Standard)

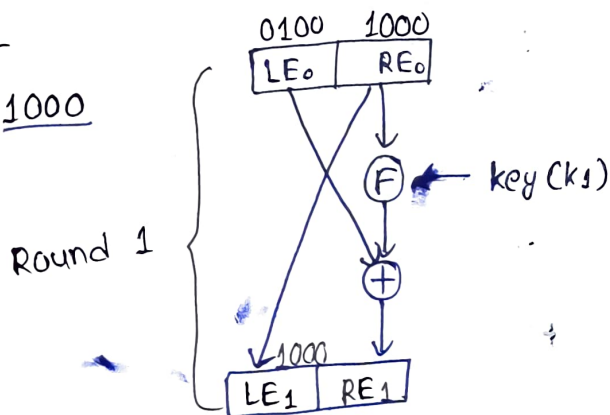
→ It is based on the Feistel cipher mathematics.

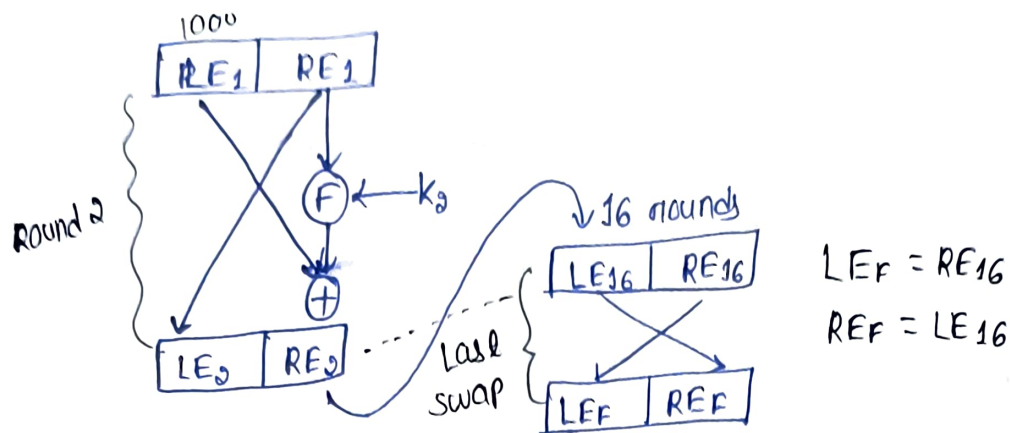
Feistel Cipher

It is a symmetric encryption structure mostly used in block ciphers where the PT is split into equal & different parts. Its structure has 16 different blocks both for encryption & decryption.

Encryption:-

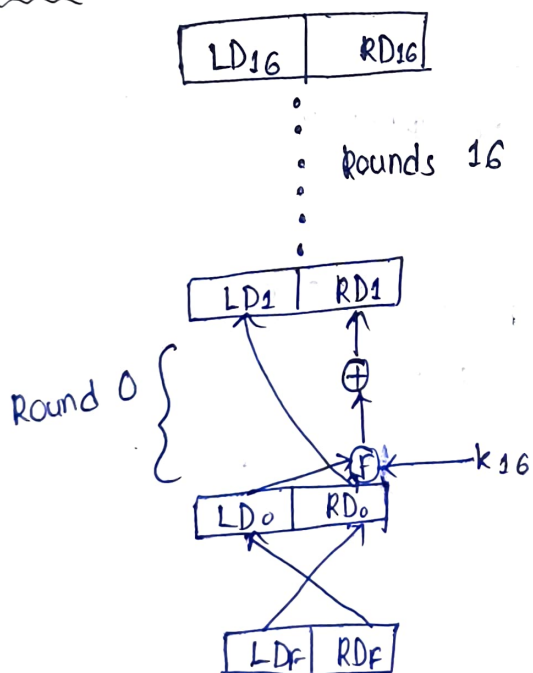
PT = 0100 1000





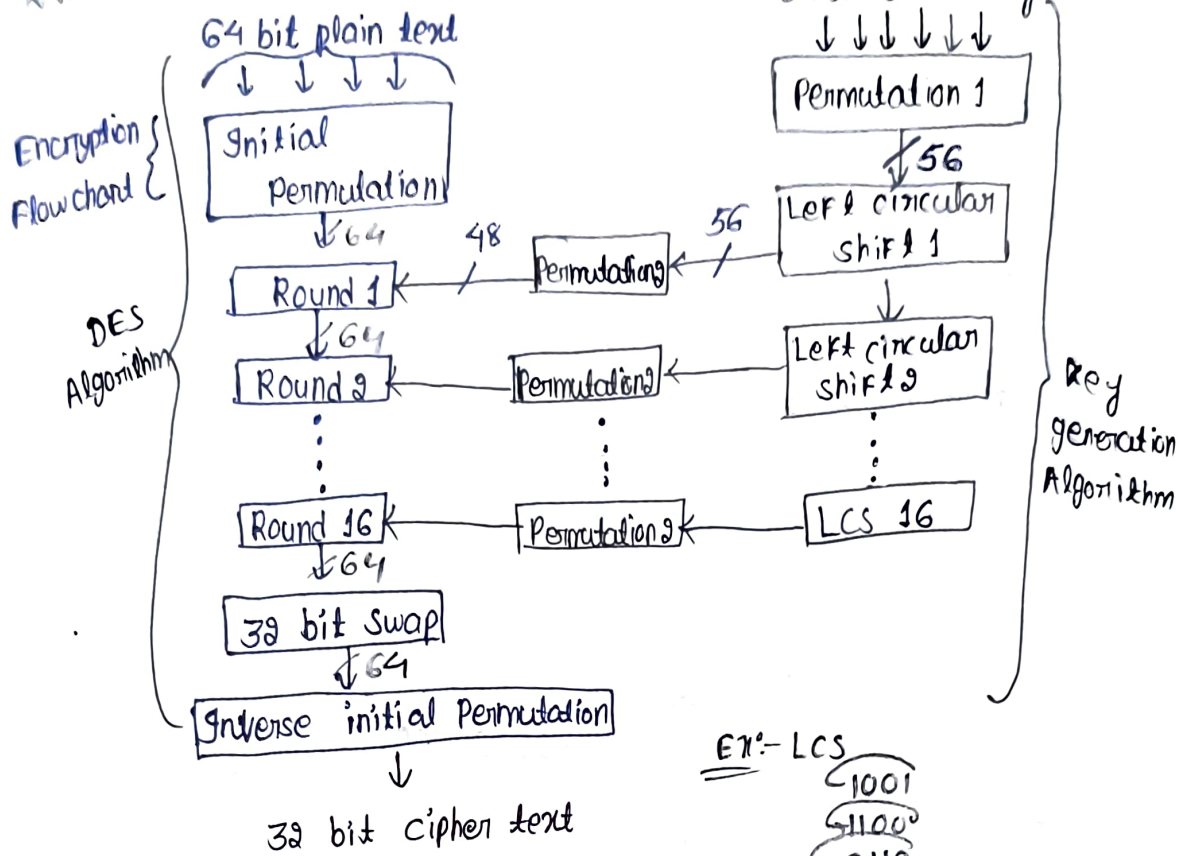
→ In encryption at the end of the last round there is a swapping operation. For decryption we need to start backwards such that round 0 of decryption starts as round 16 of encryption.

Decryption:-



	Stream cipher	Block cipher
Length	1 bit	64 or 128 bits
Design	Simple	Complex
Speed	Faster	Slower
Operating principle	Confusion	Confusion and diffusion

Data Encryption Standard:



- DES Algorithm is a block ciphering technique where the size of input PT can be 64-bits or 128-bit.
- The algorithm has 2 parts, one is DES structure and the other is key generation structure.
- It is a classical technique with complex operations in multiple rounds that help in improving the security of cryptographic technique.

→ Part 1:-

Encryption Flowchart :- It is composed of 4 parts.

- Initial permutation
- Round operation
- swapping
- Inverse initial permutation

(a) Initial permutation (IP):-

(Q) Initial Permutation (IP):

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

IP
⇒

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

①

10000010 01010000
 00000000 00000000
 00000001 00000001
 10000000 10000001 (64)
 → IP

8250000001018081 → F
(Data)

1	2	3	4	5	6	7	8
1	0	0	0	0	0	1	0
0	1	0	1	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	1
1	0	0	0	0	0	0	0
1	0	0	0	0	0	0	1

⇒

0	0	0	0	0	0	1	0
0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0
1	1	0	0	0	0	0	1
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1

09090080C1000001

(b) Round Operation:-

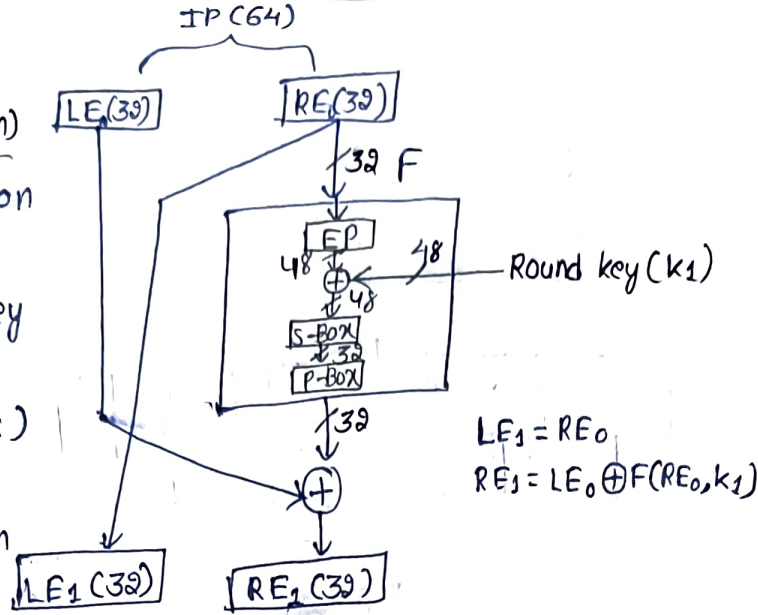
F (Mangler's Function)

→ Expansion Permutation (EP)

→ Xor with Round key

→ 8 S-Box (Substitution Box) operation

→ P-Box (Permutation Box) operation.



Expansion Permutation:-

→ It helps in converting a 32-bit data to 48-bits.

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	33

→ The EP operation is performed using a fixed expansion table that copies certain data bits to existing data bits.

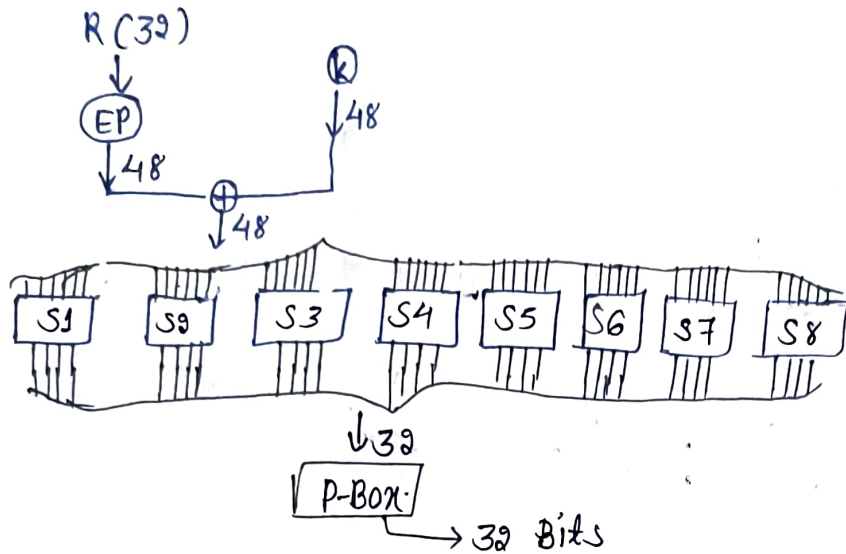
Q Convert input ~~F~~^{zero}AAFAA using EP to 48-bits.

1 1 1 1 0 0 0 0 1 0 1 0 1 0 1 0 1 1 1 1 0 0 0 0 1 0 1 0 1 0 1 0

0	1	1	1	1	0
1	0	0	0	0	1
0	1	0	1	0	1
0	1	0	1	0	1
0	1	1	1	1	0
1	0	0	0	0	1
0	1	0	1	0	1
0	1	0	1	0	1

7A15557A1555

→ S-Box (Substitution Box) :-



P-Box :-

16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

Q Calculate the output for S-box 1 if input (a) 101010
(b) 001101

(a) 101010
2nd row
5th column

→ 6 = 0110

(b) 001101
6th column
1st row

→ 13 = 1101

→ The task of S-Box is substitution and of P-Box is permutation where the 32-bit data generated by 8 S-boxes are rearranged to give 32-bit output.

Q Apply DES Algorithm for calculating the output if

PT = 10010111, key = 1010000010 if the algorithm contains

2 rounds of operation with initial permutation (IP) = $\begin{bmatrix} 2 & 6 & 8 & 1 & 4 & 8 & 5 & 7 \end{bmatrix}$,

Initial ~~IP~~ Inverse IP = 41357286, we have to use

S-Box is 081, P-box is 2,4,3,1. Expansion permutation

= $\begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$

↓

$\begin{bmatrix} 4 & 1 & 2 & 3 & 2 & 3 & 4 & 1 \end{bmatrix}$

$\begin{bmatrix} 9 & 8 & 6 \end{bmatrix}$, $P_8 = \begin{bmatrix} 6 & 3 & 7 & 4 & 8 & 5 & 1 & 0 & 9 \end{bmatrix}$

IP(8)

↓

IP

↓

$R_1 \leftarrow k_1$

↓

$R_2 \leftarrow k_2$

↓

Inv IP

↓

CT

$k = \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 10 & 10 & 0000 & 10 & (10) \end{matrix}$

↓

$P_{10} = \begin{bmatrix} 3 & 5 & 2 & 7 & 4 & 10 & 1 & 9 & 8 & 6 \end{bmatrix}$

↓

$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$

↓

00001

↓

11000

↓

$P_8 = \begin{bmatrix} 6 & 3 & 7 & 4 & 8 & 5 & 10 & 9 \end{bmatrix}$

↓

$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} (k_1)$

For key generation take $P_{10} = \begin{bmatrix} 3 & 5 & 2 & 7 & 4 & 10 & 1 & 9 \end{bmatrix}$

$\begin{matrix} S_0 & S_1 \\ 0 & 1 & 0 & 1 \\ 1 & 3 & 2 & 10 \\ 2 & 0 & 2 & 31 \\ 3 & 3 & 1 & 32 \end{matrix}$

$\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{matrix}$

$IP = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$

$R_1 = \begin{matrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 \end{matrix}$

↓

EP $\begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$

↓

$\begin{bmatrix} 4 & 1 & 2 & 3 & 2 & 3 & 4 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$

⊕ $\leftarrow k_1 (10100100)$

$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$

$S_0 \downarrow S_1$

0011 0011

↓ P-Box

$\begin{bmatrix} 2 & 4 & 3 & 1 \end{bmatrix}$

⊕ $\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} (Ans)$

Ans